Homework 4

Continuous laws, moments, and joint densities

60 points; 10 questions. Questions 7 and 10 are worth 10 points each; the others are worth 5 points each.

Due: November 3, 2026, at 10 p.m. Pacific. Final no-penalty grace cutoff: November 4, 2026, at 10 p.m. Pacific.

Handwrite your work on paper or a tablet, subject to approved accommodations. Submit one legible PDF in Gradescope and map its pages to all ten questions. Show your reasoning, support, and assumptions. Exact expressions are accepted unless stated otherwise. Write independently and disclose collaborators. Do not use AI tools to generate, solve, check, or rewrite assessed work.

Notation. CDF means cumulative distribution function. $I_{\{C\}}$ is 1 when C holds and 0 otherwise; a density with this indicator is zero outside the stated domain. Give every branch of a requested CDF, including outside the range of the variable. In $N(\mu, \sigma^2)$, the second parameter is the variance, and Φ is the standard normal CDF.

Study sequence. Start Q1–3 after October 26, Q4–6 after October 28, and Q7–10 after October 30. These are suggested stages, not additional deadlines.

1. A density and its CDF

5 points

The density of X is $f_X(x) = c(2-x)I_{\{0<x<2\}}$, where c is an unknown constant.

(a) [3] Find the value of c that makes f_X a valid density. Derive $F_X(t)$ for all real t, including the values at $t = 0$ and $t = 2$.

(b) [2] Use your CDF to find $P(1/2 < X \leq 3/2)$ and $P(X = 1)$. Explain the point probability.

2. A CDF and its density

5 points

A random variable V has CDF

$$F_V(t) = \begin{cases} 0, & t \leq 0, \\ t^2/4, & 0 < t < 2, \\ 1, & t \geq 2. \end{cases}$$

(a) [2] Find a density $f_V(v)$, including its domain, and verify that its integral over the real line is 1.

(b) [3] Calculate $f_V(3/2)$, $F_V(3/2)$, and $P(V = 3/2)$. Explain the meaning of each quantity and why a density value is not a point probability.

3. Moments and a cost measurement

5 points

The density of W is $f_W(w) = 3w^2 I_{\{0<w<1\}}$. Define a cost $C = 2 + 4W$, measured in dollars.

(a) [3] Compute $E[W]$ and $E[W^2]$ by integration, and use them to find $\mathrm{Var}(W)$.

(b) [2] Find $E[C]$, $\mathrm{Var}(C)$, and $\mathrm{SD}(C)$, with units. Explain how the coefficient 4 affects the variance.

4. A uniform waiting time 5 points

A waiting time U, measured in minutes, is uniform on $(2, 8)$.

(a) **[2]** Write the density of U and state where it is zero. Find $E[U]$ and $\mathrm{Var}(U)$, with units.

(b) **[3]** Calculate $P(U > 5)$, $P(U > 7 \mid U > 4)$, and $P(U > 3)$. Use the last two probabilities to decide whether this distribution has the memoryless property. Explain their meaning in terms of a remaining wait.

5. An exponential processing time 5 points

A processing time T is exponential with mean 5 minutes.

(a) **[3]** Find the rate λ, with units, and write the density of T. Calculate $P(T > 3)$ and $\mathrm{Var}(T)$, with units for the variance.

(b) **[2]** Calculate $P(T > 9 \mid T > 4)$ from the definition of conditional probability. Interpret the event as a remaining wait after four minutes have elapsed.

6. Probabilities under a normal model 5 points

A measurement X, in millivolts, has distribution $N(80, 36)$. Use $\Phi(1) = 0.841345$ and $\Phi(2) = 0.977250$.

(a) **[3]** Write the density of X and state its domain. Find $P(74 < X < 92)$, showing the standardized endpoints.

(b) **[2]** Find $P(|X - 80| > 12)$. Explain which measure of variability belongs in the denominator when standardizing X.

7. A joint density on a triangle 10 points

For Questions 7 and 8, X and Y have joint density $f_{X,Y}(x, y) = cI_{\{0<x<y<2\}}$, where c is an unknown constant.

(a) **[5]** Sketch the region where the density is nonzero and label its boundary lines. Find c by integration. Write the normalization integral in both orders: $dy\,dx$ and $dx\,dy$.

(b) **[5]** Find $f_X(x)$ and $f_Y(y)$, including their domains. Verify that each density integrates to 1. Explain how the joint-density region accounts for the shapes of the two marginal densities.

8. Event probabilities from a joint density 5 points

Use the joint density from Question 7.

(a) [2] Calculate $P(X > 1, Y < 1)$ and $P(X > 1)P(Y < 1)$. Use the comparison to decide whether X and Y are independent.

(b) [3] Sketch the part of the joint-density region where $Y - X > 1/2$. Write a double integral for its probability in each order, $dy\,dx$ and $dx\,dy$, and evaluate either one.

9. A joint density on a square 5 points

The joint density of R and S is $f_{R,S}(r, s) = 2r I_{\{0<r<1,\ 0<s<1\}}$.

(a) [3] Find $f_R(r)$ and $f_S(s)$, including their domains, and verify that each integrates to 1.

(b) [2] Find $P(S < R)$. Would the fraction of the square lying below the line $s = r$ give this probability? Justify your answer.

10. A service deadline 10 points

A job has two independent stage durations: $A \sim \text{Uniform}(0, 2)$ and $B \sim \text{Uniform}(0, 4)$, both measured in seconds. It misses a four-second deadline when $A + B > 4$.

(a) [5] Write the joint density of A and B, including its domain. Sketch the part of that domain where the job misses the deadline, and calculate $P(A + B > 4)$.

(b) [5] Calculate $P(A + B > 4 \mid A > 1)$. Compare it with part (a), and explain the effect of knowing that the first stage lasts more than one second.

Reading references. Harchol-Balter: pp. 134–142 (densities and continuous moments), 153–155 (joint densities), 170–175 (normal models). DeGroot–Schervish: Sections 3.2–3.5, 4.1–4.3, and the normal and exponential passages in Sections 5.6–5.7. Printed book pages; consult selected passages as needed. Neither whole chapters nor both books are required.

Optional practice

How the support determines the integration limits

Ungraded: no credit and no submission required. This is not Question 11 and is not part of the Gradescope page mapping. Homework 4 still has **10 graded questions and 60 points**.

In each model below, the joint density has the form

$$f_{X,Y}(x,y) = \begin{cases} c\,xy, & (x,y) \in S, \\ 0, & \text{otherwise.} \end{cases}$$

The expression xy is the same in every model, but the support S changes. Find a separate normalizing constant c for each distinct support. **The support determines the limits; the density determines the weight inside those limits.**

Model	Support S: all the stated conditions must hold
A	$0 < x < 1, \quad 0 < y < 1$ (the unit square).
B	$0 < x < y < 1$.
C	$0 < y < x < 1$.
D	$0 < x < 1, \quad 0 < y < 1, \quad 0 < x + y < 1$.
E	$0 < x < 1, \quad 0 < y < 1, \quad 0 < x - y < 1$.

Work through A first, then B–E. Use the same steps each time.

1. **Sketch the support.** Label the boundary lines and shade the points that satisfy every condition. Which two models describe the same region?
2. **Draw a vertical slice.** Hold x fixed. Where does y start and stop? What is the full range of x for which the slice exists? Write the limits for the order $dy\,dx$.
3. **Normalize.** Write $1 = \iint_S cxy\,dA$, substitute your limits, and solve for c. During the inner integral in y, treat x as a constant.
4. **Find $f_X(x)$.** Integrate the joint density over the allowed y values. State the range of x and give 0 elsewhere. Check that f_X integrates to 1.
5. **Find $P(X < 1/2)$.** Integrate your marginal density from 0 to 1/2. Explain why an area ratio alone need not give this probability.

A starting point for A. For any fixed $0 < x < 1$, the vertical slice runs from $y = 0$ to $y = 1$, so

$$1 = \int_0^1 \left[\int_0^1 cxy\,dy \right] dx, \qquad \int_0^1 cxy\,dy = cx \left[\frac{y^2}{2} \right]_0^1 = \frac{cx}{2}.$$

Finish the outer integral to find c, then repeat the same procedure on the other supports.

Further practice, also optional. Draw horizontal slices and write the normalization integral in the order $dx\,dy$; then find $f_Y(y)$. For each distinct model, shade the part of its support where $x + y < 1$, choose an integration order, and find $P(X + Y < 1)$. Finally, decide whether X and Y are independent. Check the support as well as the formula.

Keep the two regions separate. To normalize, integrate over the whole support. For an event probability, integrate the *same normalized density* over the part of the support satisfying that event; do not find a new c for the event.